SIMATS ENGINEERING

ASSESSMENT TOOL 1: Concept Mapping

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1522

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COURSE OUTCOME COVERED

CO1: *Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions.* (BL2)

Assessment Tool Weightage: 20%

Dave's Taxonomy: Imitation (Level 1) **Total Marks: 50**

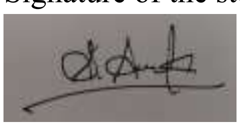
Question Count: 4

Code of Conduct

I ARAVIND KUMAR.S / 192524210 certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



Assessment Tasks

Concept Mapping Tasks

Create concept maps for the following tasks. Each map must show key nodes, link labels, and a logical hierarchy.

1. Concept Map 1: Cloud Computing Fundamentals. Include definition, characteristics, benefits, and limitations.
2. Concept Map 2: Evolution of Cloud Computing. Connect hardware evolution, internet software evolution, distributed systems, and service delivery.
3. Concept Map 3: Service Models. Map IaaS, PaaS, SaaS, and XaaS with provider responsibilities and user responsibilities.
4. Concept Map 4: Virtualization and Web Services as Enablers of Cloud. Show how these two concepts support cloud deployment and scalability.

Expected concept nodes:

- Elasticity, scalability, resource pooling, on-demand access, measured service
- Virtual machine, hypervisor, abstraction, API, SOAP, REST, provider, consumer
- Infrastructure layer, platform layer, application layer, shared responsibility

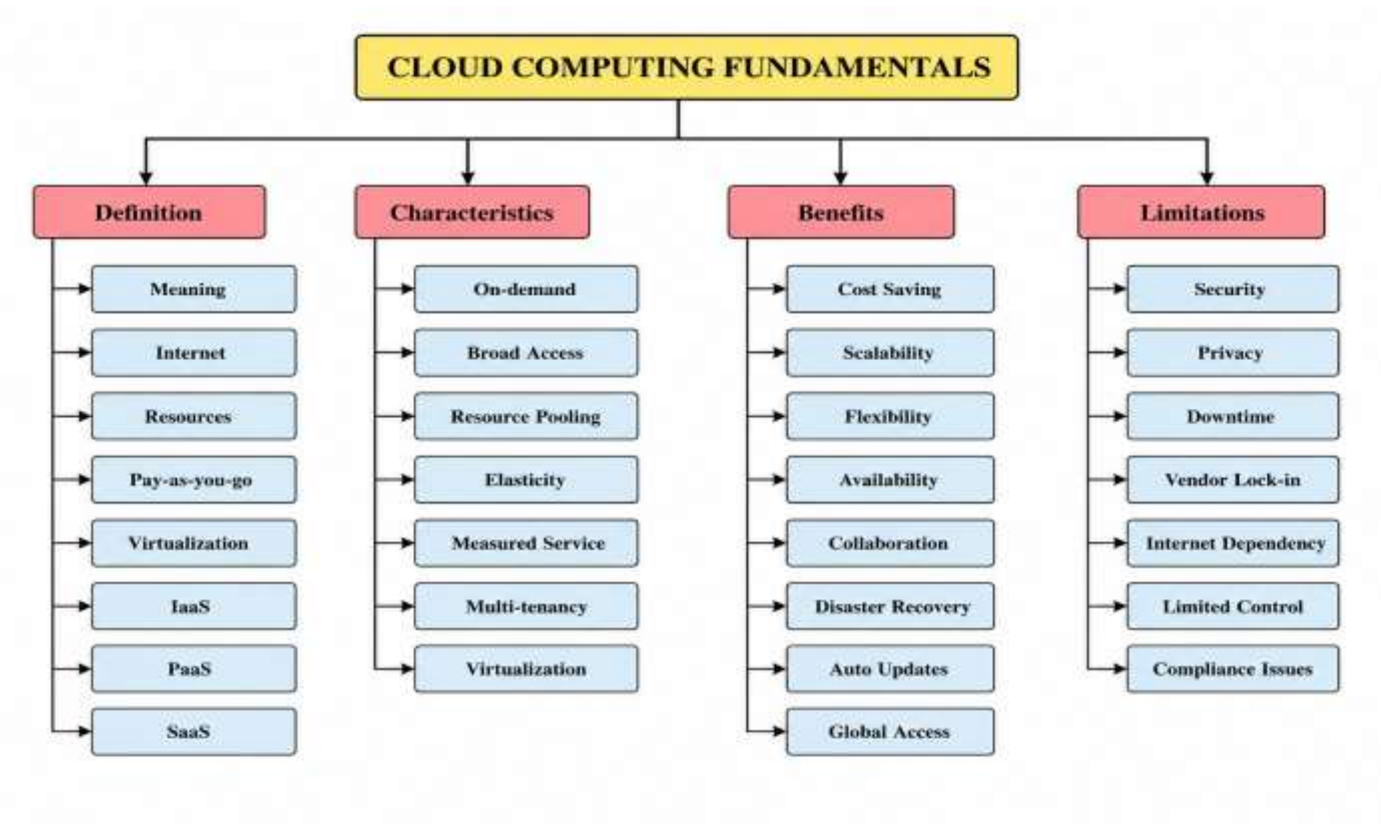
Concept Mapping Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Concept Identification	25%	All major concepts are accurately identified	Most concepts are identified correctly	Limited concepts are identified	Essential concepts are missing
Relationship Mapping	25%	Links between concepts are accurate and meaningful	Most links are correct	Some links are weak or unclear	Relationships are mostly incorrect
Hierarchy and Organization	20%	Excellent structure with clear hierarchy	Mostly organized with minor issues	Basic structure with weak hierarchy	No logical organization
Technical Relevance	20%	Map strongly reflects cloud course content	Mostly aligned to topic	Partially aligned to topic	Weak alignment to topic
Visual Clarity	10%	Neat, readable, and easy to interpret	Generally readable	Somewhat cluttered	Poorly presented

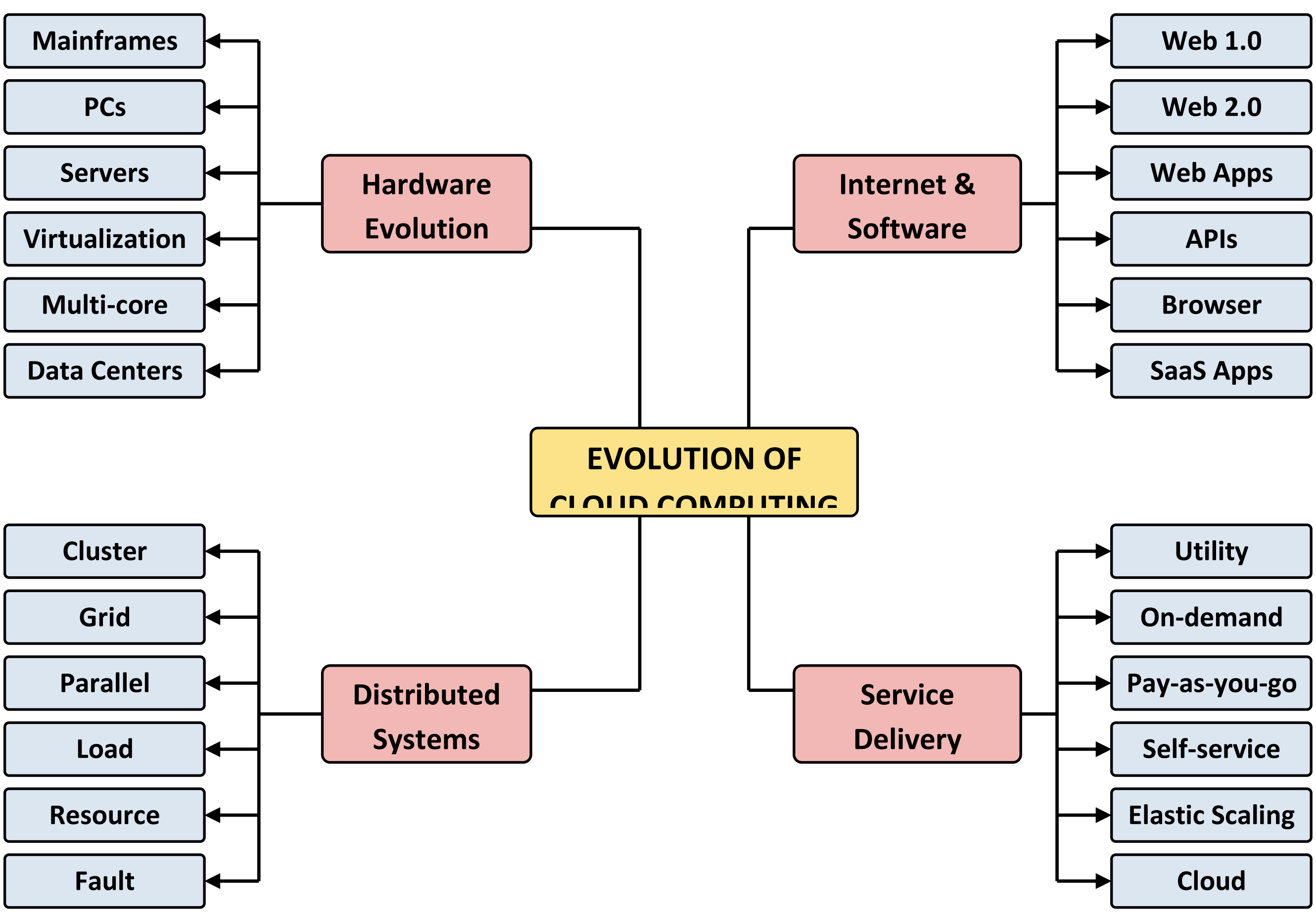
5. Questions

i). Evaluate the effectiveness of your concept map in representing cloud computing principles. Which design decisions most strongly support your solution, and why?
ii). Analyze how scalability, elasticity, and resource management influence the architecture shown in your concept map. What challenges could arise if these capabilities were absent?
iii). Justify the selection of the APIs, web services, or cloud services included in your concept map. How do they contribute to the overall functionality and efficiency of the system?
iv). Critically examine the relationships between the components in your concept map. How do these interactions improve reliability, performance, or user experience?
v). Reflect on the process of developing your concept map. How has creating and analyzing the map changed your understanding of cloud computing architecture, and what improvements would you make if you were redesigning it?

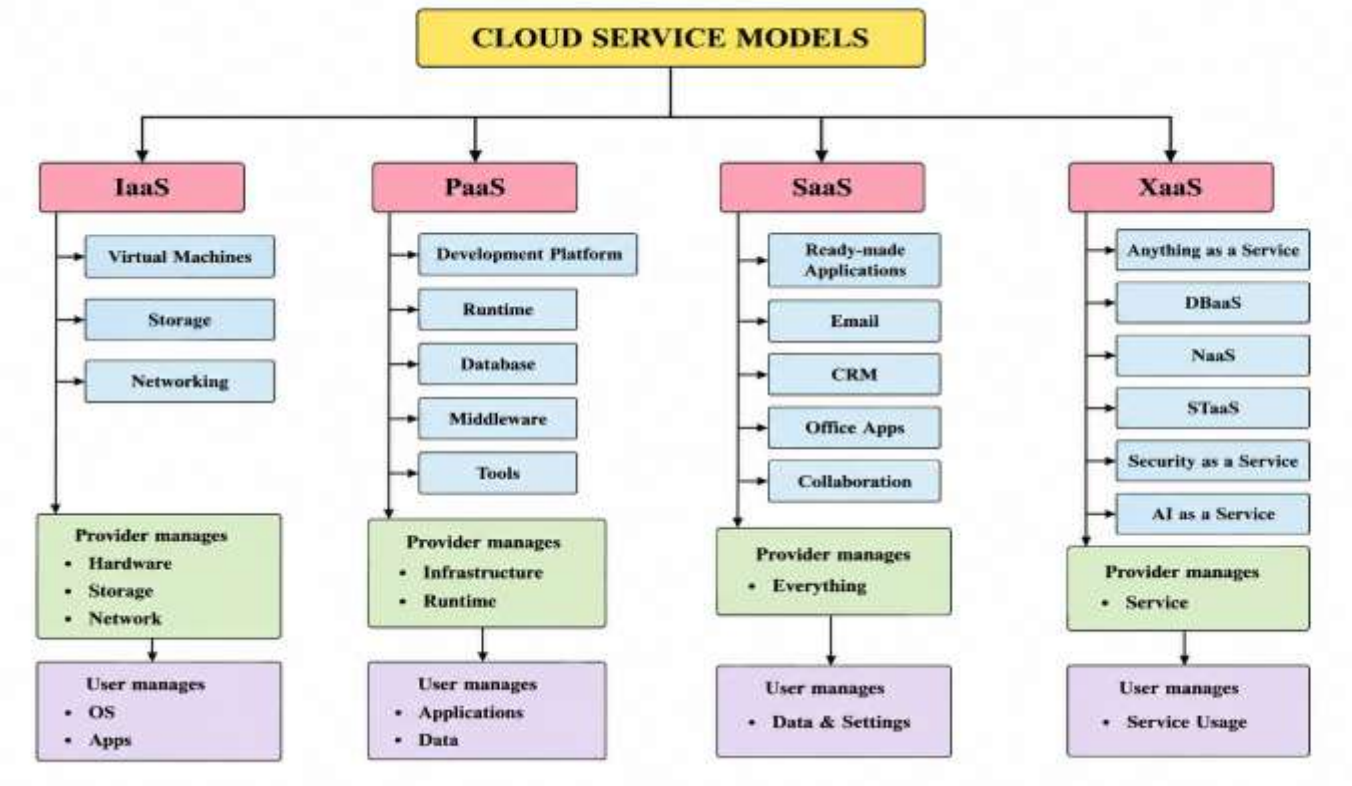
1.Cloud Computing Fundamentals



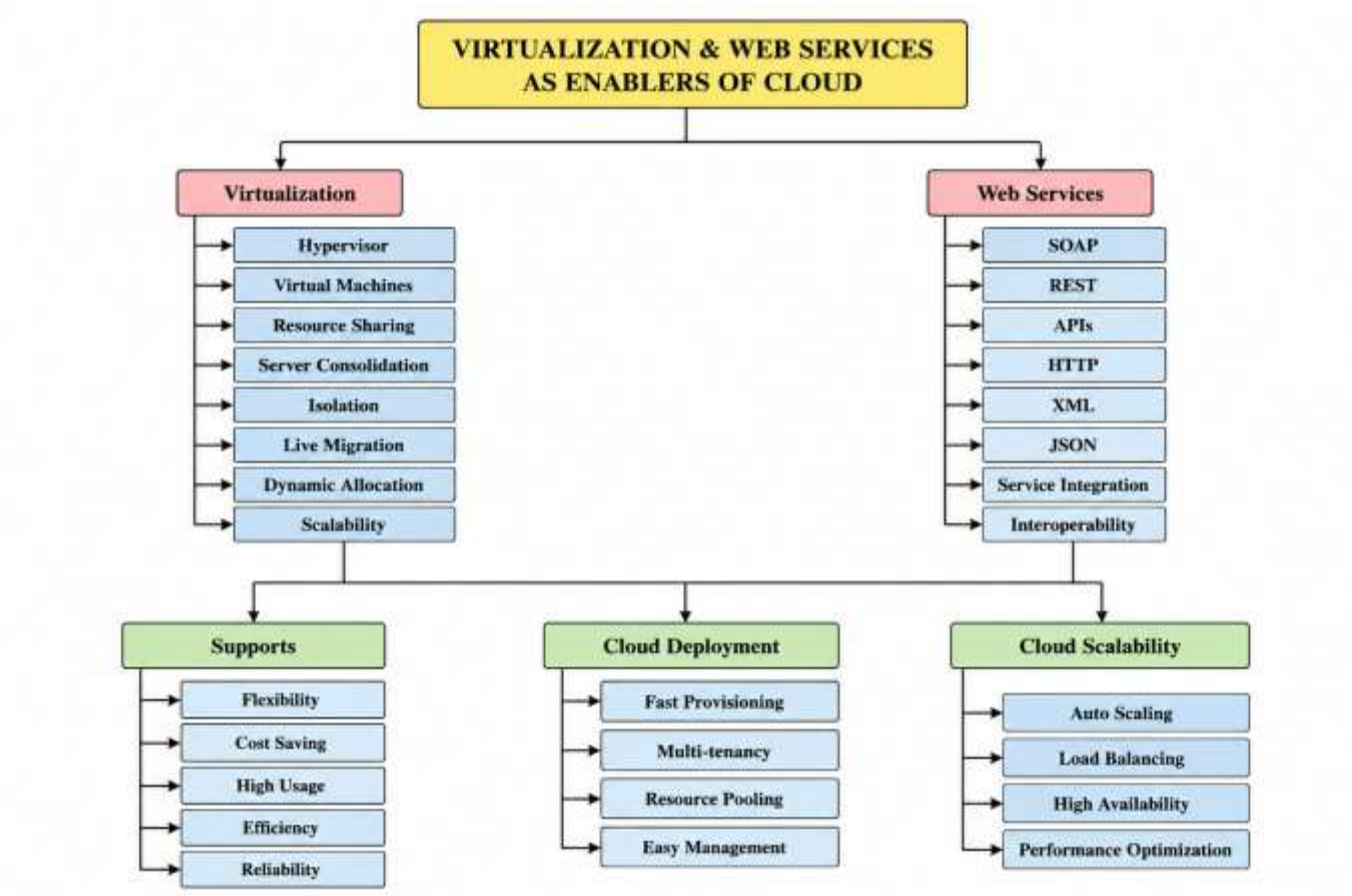
2. : Evolution of Cloud Computing



3. Service Models.



4.Virtualization and Web Services as Enablers of Cloud



i) Evaluate the effectiveness of your concept map.

Answer :

- Represents cloud computing concepts in a clear and organized manner.
- Uses a hierarchical structure for easy understanding.
- Connects related concepts using branches and sub-branches.
- Covers key topics such as characteristics, service models, deployment models, benefits, and limitations.
- Improves learning by simplifying complex concepts.
- Design decisions like logical grouping and visual connections make the concept map more effective.

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ii) Analyze how scalability, elasticity, and resource management influence the architecture.

Answer :

- Scalability: Increases or decreases resources based on workload.
- Elasticity: Automatically allocates and releases resources when demand changes.
- Resource Management: Optimizes the use of CPU, memory, storage, and network resources.
- Improves system performance and availability.
- Reduces operational costs.
- Without these capabilities:
    - Slow system performance.
    - Resource shortages during high traffic.
    - Increased costs.
    - Poor user experience.

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iii) Justify the selection of APIs, web services, or cloud services.

Answer :

- APIs enable communication between different applications.
- Web services support data exchange over the internet.
- IaaS provides virtual infrastructure.
- PaaS offers a platform for application development.
- SaaS delivers software through the internet.
- These services improve flexibility, integration, scalability, and overall efficiency.

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iv) Critically examine the relationships between the components.

Answer :

- Virtualization supports cloud infrastructure.
- Service models depend on cloud infrastructure.
- Deployment models determine where cloud services are hosted.
- Security protects cloud resources and user data.
- Scalability and resource management improve system efficiency.
- These interactions provide:
    - Higher reliability.
    - Better performance.
    - Faster response time.
    - Improved user experience.

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v) Reflect on the process of developing your concept map.

Answer :

- Improved understanding of cloud computing architecture.
- Helped visualize the relationships between cloud concepts.
- Made learning easier through structured organization.
- Enhanced knowledge of service and deployment models.
- If redesigned:
    - Add real-world examples (AWS, Azure, Google Cloud).
    - Use more colors and icons for better visualization.
    - Include more details on security and virtualization.
    - Improve layout for easier readability.